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Hybrid AI-based Identity and Age Verification System

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Advanced Machine Learning
Intelligent Systems

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Contents

Abstract **2**

1 Introduction **2**
1.1 Context and Motivation 2
1.2 The problem to solve 2
1.3 Main contribution 2

2 State of the Art **3**
2.1 Current Age Verification Systems 3
2.2 Liveness Detection and Voice Biometrics 3
2.3 Positioning and Differentiation of this Work 3

3 Preambles (Technical Foundations) **4**
3.1 Residual Architectures (ResNet) 4
3.2 Multi-task Learning 4
3.3 Visual Interpretability (Grad-CAM) 4
3.4 Speech Recognition (ASR), Textual Similarity and TTS 4
3.5 Metaheuristics: Genetic Algorithms 5

4 Technical Proposal **5**
4.1 Computer Vision (CV) 5
4.2 Speech and Natural Language Processing (NLP) 6
4.3 Advanced Machine Learning (Advanced ML) 6
4.4 Intelligent Systems (IS) 6

5 UI and UX Design (Gradio) **7**

6 Evaluation and Results **8**
6.1 Training Dataset 8
6.2 Quantitative Analysis 8
6.3 Training Challenges and Qualitative Analysis 8

7 Conclusions **9**

8 Future Work **9**

Summary

This document details the development of a comprehensive identity verification application based on a multi-factor approach. The system combines voice recognition to ensure the user's physical presence with computer vision algorithms for age estimation. Through the use of deep learning architectures (ResNet18), natural language processing techniques, and metaheuristic optimization (Genetic Algorithms), a robust, scalable system with a mid-level user interface has been achieved. The results demonstrate a 97.66% accuracy in adulthood classification.

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1 Introduction

1.1 Context and Motivation

With the rise of the digitalization of sensitive services (banking, adult content, gambling) and the new intention to implement adulthood regulations for the use of social networks, the need to verify the user's identity remotely has become imperative. Traditional single-factor systems are vulnerable to spoofing attacks using static photos or recordings. This project proposes a hybrid solution that links behavioral biometrics (voice) with physical biometrics (the face).

1.2 The problem to solve

The main challenge consists of creating a system that:

1. Verifies the user's real presence (not a photo or a bot).
2. Estimates age with a minimum margin of error using limited hardware (a standard laptop).
3. Is interpretable, trying to allow understanding of why the AI makes certain decisions.

1.3 Main contribution

The main innovation lies in the **sequential verification chain** optimized through **Intelligent Systems**. The system does not allow advancing to age estimation if the voice challenge has not been successfully passed, drastically reducing the attack surface for automated fraud.

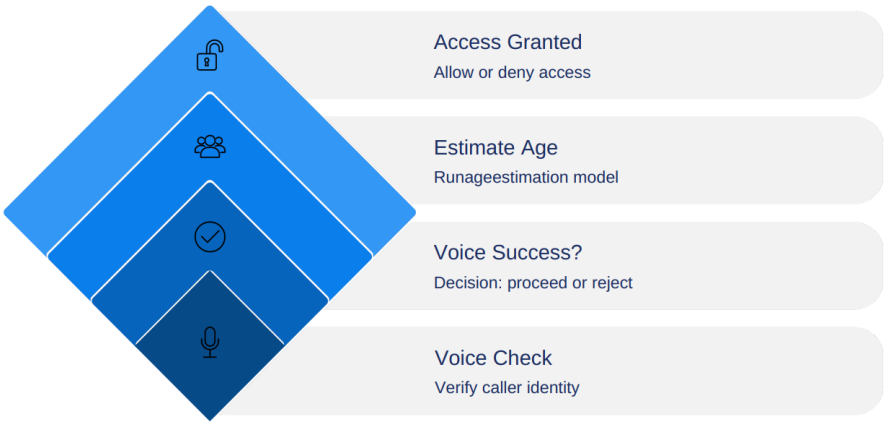

Figure 1: General flow diagram

2 State of the Art

2.1 Current Age Verification Systems

Currently, the vast majority of age verification systems on the Internet are based on obsolete or insecure methods, such as self-declaration (the classic "I am over 18" button) or static validation of identity documents. Although the industry has recently started to adopt computer vision to estimate age from facial images, these single-modal systems (based only on image) present a critical weakness: they are highly vulnerable to presentation attacks, where an attacker can fool the system by showing a printed photograph or a screen with an adult's face.

2.2 Liveness Detection and Voice Biometrics

To combat this fraud, the state of the art proposes "Proof of Life" (*Liveness Detection*) techniques. While visual approaches usually ask the user to blink or smile, behavioral biometrics based on voice have proven to be a much more robust barrier. Forcing a user to read a dynamic and random text requires a cognitive and physical response in real time. This speech and natural language processing technique guarantees human presence, but by itself does not serve to determine the exact demographics or age of the user.

2.3 Positioning and Differentiation of this Work

The big gap in the current state of the art is that systems blindly trust a static image, or they use multimodal approaches that turn out to be computationally very heavy "black boxes".

The main difference and contribution of this project is its sequential hybrid architecture. Instead of simply analyzing a face in isolation, the proposed system first requires passing a dynamic voice challenge (reading a random phrase generated on the spot). This first layer acts as an anti-fraud filter for bots or static photos. Only when the natural language processing module certifies that there is a real human interacting live, the second phase is enabled: age estimation through computer vision. This separation of responsibilities (voice to certify presence,

image to calculate age) achieves a level of security typical of complex multimodal systems, while maintaining a lightweight, logical, and interpretable execution.

3 Preambles (Technical Foundations)

To understand the proposed hybrid architecture, it is necessary to briefly define the algorithmic paradigms and mathematical models on which the system is based.

3.1 Residual Architectures (ResNet)

Classic Convolutional Neural Networks (CNN) suffer from the vanishing gradient problem as their depth increases. The ResNet (*Residual Network*) architecture solves this by introducing "shortcut connections" (*skip connections*) that allow the signal to skip certain layers. This facilitates the training of deep networks capable of extracting complex facial features (such as wrinkles or skin texture) in a highly efficient way.

3.2 Multi-task Learning

Multi-task learning is a Machine Learning paradigm where a single model learns to solve several problems simultaneously, sharing internal representations. Instead of having one network to classify (adult/minor) and another to estimate the exact age, a multi-task network forks its final layers into several "heads". Mathematically, it requires minimizing a combined loss function that weights the error of each individual task (for example, combining *Binary Cross Entropy* and *Mean Absolute Error*).

3.3 Visual Interpretability (Grad-CAM)

Grad-CAM (*Gradient-weighted Class Activation Mapping*) is an explanatory technique for convolutional networks. It works by calculating the gradients of the predicted class with respect to the feature maps of the last convolutional layer. This generates a "heatmap" that spatially highlights which pixels of the original image had the greatest weight in the network's decision.

3.4 Speech Recognition (ASR), Textual Similarity and TTS

The processing of the voice challenge requires several phases to achieve a natural interaction. First, Automatic Speech Recognition (ASR) systems transform what is said into text. Once transcribed, it is necessary to evaluate the reading accuracy using sequence similarity algorithms (such as the Ratcliff/Obershelp pattern matching or the Levenshtein distance), which mathematically calculate the overlap ratio between the expected and detected text string. Finally, to guide the user and close the communication loop, Text-to-Speech (TTS) systems are used. These technologies reverse the ASR process, converting dynamic text into audio to provide real-time, accessible sound feedback.

3.5 Metaheuristics: Genetic Algorithms

Compared to brute-force methods (*Grid Search*) which are computationally impossible for hyper-parameter search in Deep Learning, Genetic Algorithms explore the solution space stochastically. Based on evolutionary theory, they evaluate a "population" of configurations using an adaptation function (*fitness*) and generate new generations through probabilistic selection, crossover, and mutation operators, rapidly converging to high-quality global or local optima.

4 Technical Proposal

The system is divided into four main modules, each representing an advanced area of study:

4.1 Computer Vision (CV)

The visual core uses a **ResNet18** pre-trained with ImageNet.

- **Adaptation Architecture**: The final layer of the ResNet was removed and replaced by a sequence of dense layers (`nn.Linear`) with ReLU activations and Dropout for high-level feature extraction.
- **Interpretability with Grad-CAM**: To comply with "Explainable AI" standards, the Grad-CAM algorithm was implemented. This calculates the gradients of the last convolutional layer (`layer4`) with respect to the output to generate a heatmap overlaid on the user's photo.

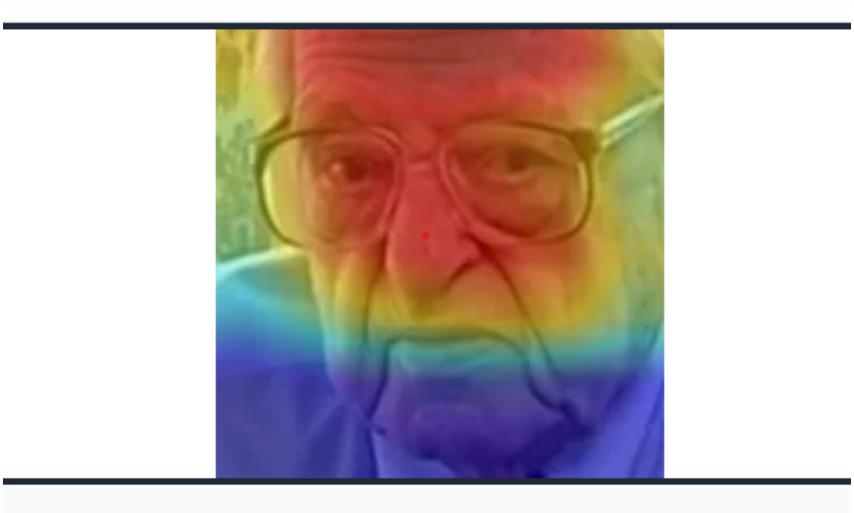

Figure 2: Example of Grad-CAM result showing how the AI focuses on the eyes and forehead area to predict age

4.2 Speech and Natural Language Processing (NLP)

The anti-fraud module works through a reading challenge:

1. **Challenge Generation**: The system selects a random basic-level phrase (e.g. *"Hello my name is John"*).
2. **Capture and Transcription**: The `SpeechRecognition` library is used to send the audio to Google's servers and obtain a real-time transcription.
3. **Semantic Verification**: Normalization is applied (removal of punctuation, capital letters, and special characters) and similarity is calculated using `SequenceMatcher`.
4. **Voice Feedback (TTS)**: A voice guide was implemented using `gTTS` which converts dynamic instructions into temporary `.mp3` files, improving interactivity.

4.3 Advanced Machine Learning (Advanced ML)

The model has been designed under the **Multi-task Learning** paradigm.

- **Dual Output**: The network has two final heads:
  - **Head A (Classification)**: Sigmoid output to determine if the user is an adult (Adult/Minor).
  - **Head B (Regression)**: Linear output to estimate exact age.
- **Combined Loss Function**: The training minimizes a weighted loss:

$$\text{Loss} = 0.7 \times \text{BCE (Classification)} + 0.3 \times \text{MAE (Regression)}$$

- **Robustness**: An aggressive **Data Augmentation** policy was applied, including random crops (`RandomResizedCrop`), color variations, and random erasing of areas (`RandomErasing`) to force the network not to depend on spurious background correlations.

4.4 Intelligent Systems (IS)

A **Genetic Algorithm** was implemented to automate the search for the best hyperparameters.

- **Search Space**:
  - `learning_rate`: [0.01, 0.001, 0.0001]
  - `batch_size`: [16, 32, 64]
  - `dropout`: [0.2, 0.3, 0.5]
- **Evolution**:
  - **Selection**: Tournament method (selects the best of N random individuals).
  - **Crossover**: Random mix of the parameters of two parents to create a child.
  - **Mutation**: 20% probability that a parameter changes randomly to avoid local minima.

5 UI and UX Design (Gradio)

The interface was developed with **Gradio**, prioritizing a "Premium" and minimalist aesthetics.

- **Custom CSS**: A centralized "card" design with modern styles (soft borders, subtle shadows) was implemented.
- **Sequential Flow**: The interface logically hides future steps. "Step 2" is only visible once the voice has been verified.
- **Compatibility**: The code has been updated to comply with **Gradio 6.0** standards, guaranteeing optimal performance in modern browsers.

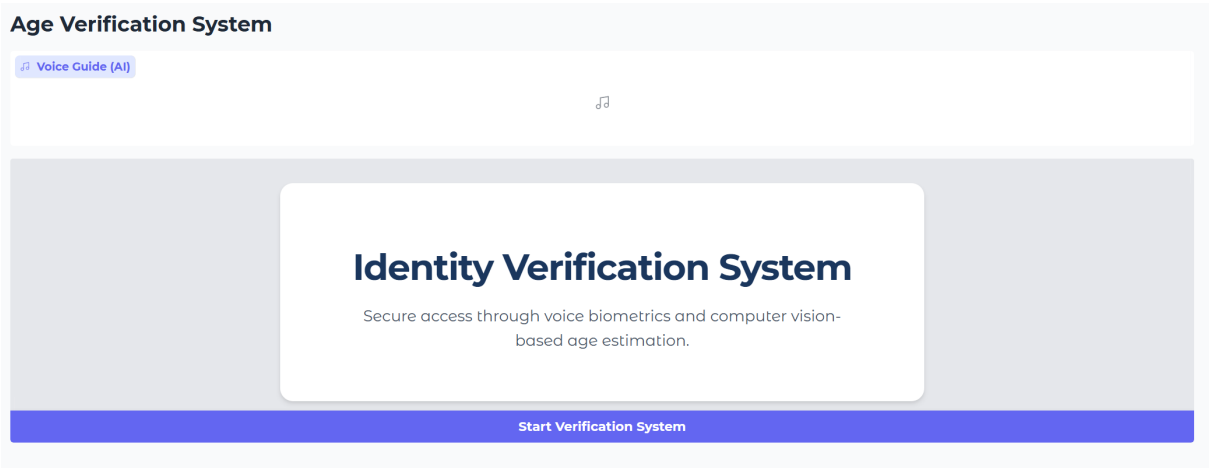

Figure 3: Capture of the welcome screen with the centered design

6 Evaluation and Results

6.1 Training Dataset

The **UTKFace** dataset was used, composed of more than 20,000 images with great diversity. An 80% split for training and 20% for validation was performed.

6.2 Quantitative Analysis

After optimization with the Genetic Algorithm, the results on the validation set were:

Metric	**Obtained Value**
Accuracy (Adult/Minor Classification)	97.66%
MAE (Mean Error in Estimated Age)	4.93 years
F1-Score (Classification)	98.50%

Table 1: Quantitative results of the model after optimization.

This demonstrates that the system is reliable for distinguishing between adults and minors, with a very contained age error given that it is a lightweight model (ResNet18).

6.3 Training Challenges and Qualitative Analysis

The greatest technical challenge of the project was the training phase of the visual model, which initially failed a lot and suffered from *overfitting*, memorizing the training images and failing in validation. To solve it, an aggressive data augmentation policy was implemented that included crops, color variations, and the `RandomErasing` technique (hiding parts of the face with black squares), forcing the network to generalize. This, added to the tuning of the *Dropout* layers using the genetic algorithm, stabilized the learning.

On the other hand, an imbalance was detected in the original dataset, which contained very few images of people over 70 years old, which skyrocketed the regression error (MAE) in those ages. The implemented solution was the weighted combined loss function: by giving a 70% weight to binary classification versus 30% to regression, the model prioritized getting right whether the person was of legal age (where it reached 97.66%), mitigating the impact of the 70-year bias on the final access decision.

Finally, the qualitative analysis with Grad-CAM revealed that, although the model usually focuses correctly on key facial features (eyes, corners of the mouth), it occasionally activated corners or areas of the image background, which underlines the importance of the spatial data augmentation applied to subtract weight from the context.

7 Conclusions

The development of this system demonstrates the feasibility of implementing robust age verification by combining sequential Artificial Intelligence methods on conventional hardware.

The main contributions achieved are:

1. **Hybrid Security Architecture:** A sequential funnel has been successfully created where the visual estimation is only executed if the natural language processing module (`speech_nlp`) certifies a similarity greater than 75% in reading the random phrase, with the possibility of changing the requirement.

2. **Balanced Multi-task Model:** The modified ResNet18 network managed to process two simultaneous tasks through a combined loss function ($\alpha = 0.7$ for classification and 0.3 for regression). This allowed reaching a 97.66% accuracy to distinguish adults from minors, and a Mean Absolute Error (MAE) of 4.93 years in the numerical estimation.

3. **Transparency and Explainability:** The integration of Grad-CAM provides explainability to the end user, visually demonstrating, through heatmaps, which spatial regions of the face determine the CNN's output; even though it sometimes seems to focus on meaningless corners of the image, most of the time it does so correctly.

4. **Automated Optimization:** The design of the custom Genetic Algorithm demonstrated that the stochastic search for hyperparameters (Learning Rate, Batch Size, Dropout) is a viable and efficient alternative to computationally expensive brute-force methods.

Ultimately, the combination of behavioral (voice) and physical (face) biometrics results in a system that is more resistant to static spoofing attacks than traditional single-modal approaches.

8 Future Work

Despite the good results, the project presents certain technical limitations that open up avenues for research and improvement:

- **Own Application Development:** Currently, the user interface is orchestrated and deployed as a web prototype using the Gradio library (`app.py`). For a production environment, the development of a native application (mobile or full web) with a dedicated client-server architecture is contemplated, which would improve latency, interface personalization, and deployment security.

- **Documentary Backup System:** Since no biometric model has 100% accuracy right now, there is a risk of false negatives (classifying an adult as a minor). It is proposed to add an alternative verification route that allows the user to upload a photograph of their National Identity Document (DNI) to apply OCR (Optical Character Recognition) techniques as a last resort if facial estimation fails, in addition to also serving as identity verification of the person, comparing the uploaded image with the ID document image.

- **Dependence on External APIs:** Currently, the voice verification module depends on Google's recognition API. A limitation of this is the need for a permanent internet connection and possible latencies. As future work, it is proposed to replace this call with a local inference *Speech-to-Text* model (such as Whisper) to make the system completely autonomous, since right now the tests carried out with Whisper did not yield good results.

- **Fine Tuning of the Similarity Function:** Anti-fraud validation uses `SequenceMatcher` based on character sequence similarity. A user with a strong accent could legitimately fail the test. The implementation of semantic or phonetic similarity that is more tolerant of pronunciation variations could be investigated, instead of strict text similarity.

- **Dataset Expansion and Transfer Learning:** The model was trained with UTKFace, which has cropped and centered images. Although aggressive *Data Augmentation* (`RandomErasing`, crops) was applied, the system could benefit from performing *Transfer Learning* on datasets captured directly in web environments (webcam-type images) to reduce the gap between the training domain and the Gradio application's inference domain. Furthermore, the dataset did not contain many images from 70 years old onwards, which is why it failed a lot in older ages, and it would be good to expand it.

- **Integration of the Genetic Algorithm in the Architecture:** Currently, the evolutionary module for hyperparameter search (`genetic.py`) is executed independently of the main deployment loop. A technical improvement would be to build an automated *pipeline* where the system can be periodically retrained and self-optimized as it receives new input data.

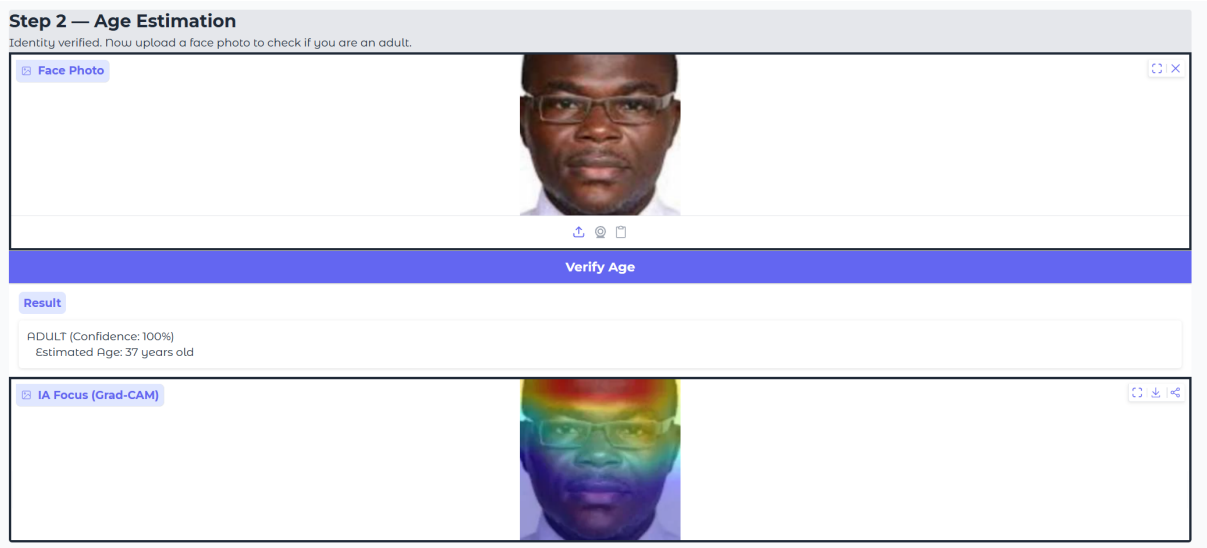

Figure 4: Capture of the final system, in this case predicting 37 and being 36 years old